



Hypothesis Testing · Mark Scheme

Statistics Chapter 7: Hypothesis Testing. M = method mark A = accuracy mark (depends on M) B = independent mark
ft = follow through

Section A – The language of hypothesis testing

- 1 (a) The assumption that the parameter takes a particular value, assumed true unless there is evidence against it *B1*.
(b) What we are looking for evidence of: how the parameter differs if H_0 is wrong *B1*.
- 2 One-tailed: tests for an increase *or* a decrease only *B1*; two-tailed: tests for a change in either direction *B1*.
- 3 (a) The set of values of the test statistic that lead to rejecting H_0 *B1*. (b) The probability chosen as the threshold for rejecting H_0 *B1*. (c) The actual probability of rejecting H_0 when it is true (the probability of the critical region) *B1*.
- 4 (a) $H_0: p = 0.5, H_1: p > 0.5$ *B1*; (b) $H_0: p = \frac{1}{6}, H_1: p < \frac{1}{6}$ *B1*; (c) $H_0: p = 0.1, H_1: p \neq 0.1$ *B1*.
- 5(a) $H_0: p = 0.5, H_1: p > 0.5$ *B1*.
- 5(b) $X \sim B(12, 0.5)$ *M1*; $P(X \geq 10) = 0.0193, P(X \geq 9) = 0.0730$ *M1*; critical region $X \geq 10$ *A1*.
- 5(c) 0.0193 (1.93%) *B1*.
- 5(d) 9 is not in the critical region: insufficient evidence that the coin is biased towards heads *B1*.

Section B – Binomial hypothesis tests

- 6 $H_0: p = 0.3, H_1: p > 0.3, X \sim B(10, 0.3)$ *B1*; $P(X \geq 6) = 0.0473 < 0.05$ *M1*; reject H_0 : evidence the proportion buying a drink has increased *A1*.
- 7 $H_0: p = 0.5, H_1: p < 0.5$ *B1*; $P(X \leq 5) = 0.0207 < 0.05$ *M1*; reject H_0 : evidence the probability of red is less than 0.5 *A1*.
- 8 $X \sim B(20, 0.25)$ *M1*; $P(X \geq 9) = 0.0409$ *A1*; $P(X \geq 8) = 0.1018$, so CR is $X \geq 9$ *A1*; actual significance level 0.0409 *B1*.
- 9 $X \sim B(20, 0.4)$; $P(X \leq 3) = 0.0160, P(X \leq 4) = 0.0510$ *M1*; lower tail $X \leq 3$ *A1*; $P(X \geq 13) = 0.0210, P(X \geq 12) = 0.0565$ *M1*; upper tail $X \geq 13$ *A1*; actual significance level $0.0160 + 0.0210 = 0.0370$ *B1*.
- 10(a) p = proportion of laptops returned *B1*; $H_0: p = 0.1, H_1: p > 0.1$ *B1*.
- 10(b) $X \sim B(30, 0.1)$ *M1*; $P(X \geq 7) = 1 - P(X \leq 6) = 0.0258$ *M1 A1*; $0.0258 < 0.05$: reject H_0 , there is evidence that the proportion of laptops returned has increased *A1*.
- 10(c) $P(X \geq 6) = 0.0732 > 0.05$ *M1*; $P(X \geq 7) = 0.0258 < 0.05$ *A1*; critical region $X \geq 7$ *A1*.
- 10(d) 0.0258 (2.58%) *B1*.

Section C – Testing the mean of a normal distribution

- 11 $\bar{X} \sim N\left(50, \frac{4^2}{16}\right), z = \frac{51.6 - 50}{1} = 1.6$ *M1 A1*; $1.6 < 1.6449$: do not reject H_0 , insufficient evidence that $\mu > 50$ *A1*.
- 12 $z = \frac{36 - 40}{6/3} = -2$ *M1 A1*; $-2 > -2.3263$: do not reject H_0 , insufficient evidence at 1% that $\mu < 40$ *A1*.
- 13 $H_0: \mu = 100, H_1: \mu \neq 100$ *B1*; $z = \frac{102 - 100}{5/5} = 2$ *M1*; critical value 1.96 *B1*; $2 > 1.96$: reject H_0 , evidence the mean mass is not 100 g *A1*.
- 14 $\bar{X} \sim N\left(40, \frac{64}{20}\right)$ *B1*; $\frac{c - 40}{8/\sqrt{20}} = 1.6449$ *M1 B1*; critical region $\bar{X} > 42.94$ *A1*.
- 15(a) $H_0: \mu = 48, H_1: \mu < 48$ *B1*; $\bar{X} \sim N\left(48, \frac{3.2^2}{40}\right)$ *B1*.



15(b) $z = \frac{47.1 - 48}{3.2/\sqrt{40}}$ $M1$; $= -1.779$ $A1$; critical value -1.6449 $B1$; $-1.779 < -1.6449$, reject H_0 $M1$; there is evidence that the mean lifetime has decreased $A1$.

15(c) $p\text{-value} = P(Z < -1.779) = 0.0376$ $B1$; reject at 4% (or any level above 3.76%) $B1$.

Section D – Testing for correlation

16 $H_0: \rho = 0$, $H_1: \rho > 0$ $B1$, where ρ is the population correlation coefficient $B1$.

17 $H_0: \rho = 0$, $H_1: \rho > 0$ $B1$; critical value 0.5494 $B1$; $0.61 > 0.5494$: reject H_0 , evidence of positive correlation $B1$.

18 $H_1: \rho \neq 0$, 2.5% in each tail $B1$; critical value 0.5140 $B1$; $|-0.48| < 0.5140$: do not reject H_0 , insufficient evidence of correlation $B1$.

19 (a) $0.42 < 0.5155$: no evidence of positive correlation at 1% $B1$ $B1$; (b) $0.42 > 0.3783$: evidence of positive correlation at 5% $B1$.

20(a) $H_0: \rho = 0$ $B1$; $H_1: \rho > 0$ $B1$.

20(b) Critical value 0.4973 $B1$; $0.55 > 0.4973$, reject H_0 $M1$; there is evidence of positive correlation between daily total sunshine and daily maximum temperature at Camborne $A1$.

20(c) The 1% critical value is 0.6581 and $0.55 < 0.6581$ $B1$.

20(d) Only 12 days were used, a small sample $B1$; the LDS only covers May to October, so the results may not apply to winter months $B1$.

Roar-level challenge

21 $H_0: p = 0.35$, $H_1: p < 0.35$, $X \sim B(30, 0.35)$ $B1$; $P(X \leq 4) = 0.0075$, $P(X \leq 5) = 0.0233$ $M1$; critical region $X \leq 4$ $A1$; actual significance level 0.0075 $B1$.